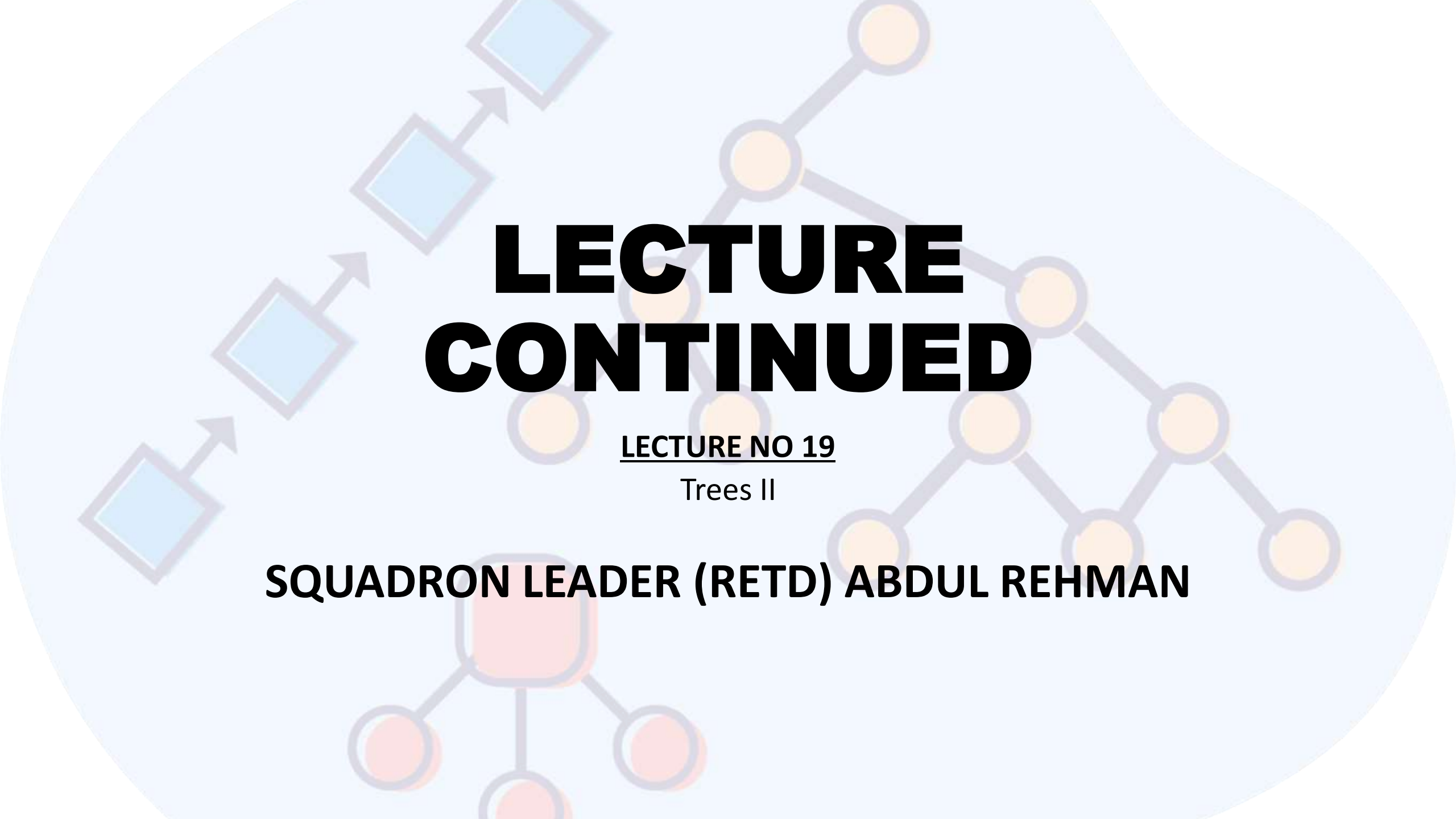




LECTURE CONTINUED

LECTURE NO 19

Trees II

SQUADRON LEADER (RETD) ABDUL REHMAN

PROPERTIES OF BINARY TREES

- **Maximum Nodes**

- A binary tree of height h has at most $2^{(h+1)} - 1$ nodes

- **Minimum Height**

- A binary tree with n nodes has a minimum height of $\lceil \log_2(n + 1) \rceil - 1$
- A binary tree with n nodes has at least $\left\lceil \frac{n}{2} \right\rceil$ leaf nodes

SIZE, DEPTH AND HEIGHT

- **Size**

- The size of a tree is the total number of nodes present in the tree

- **Depth**

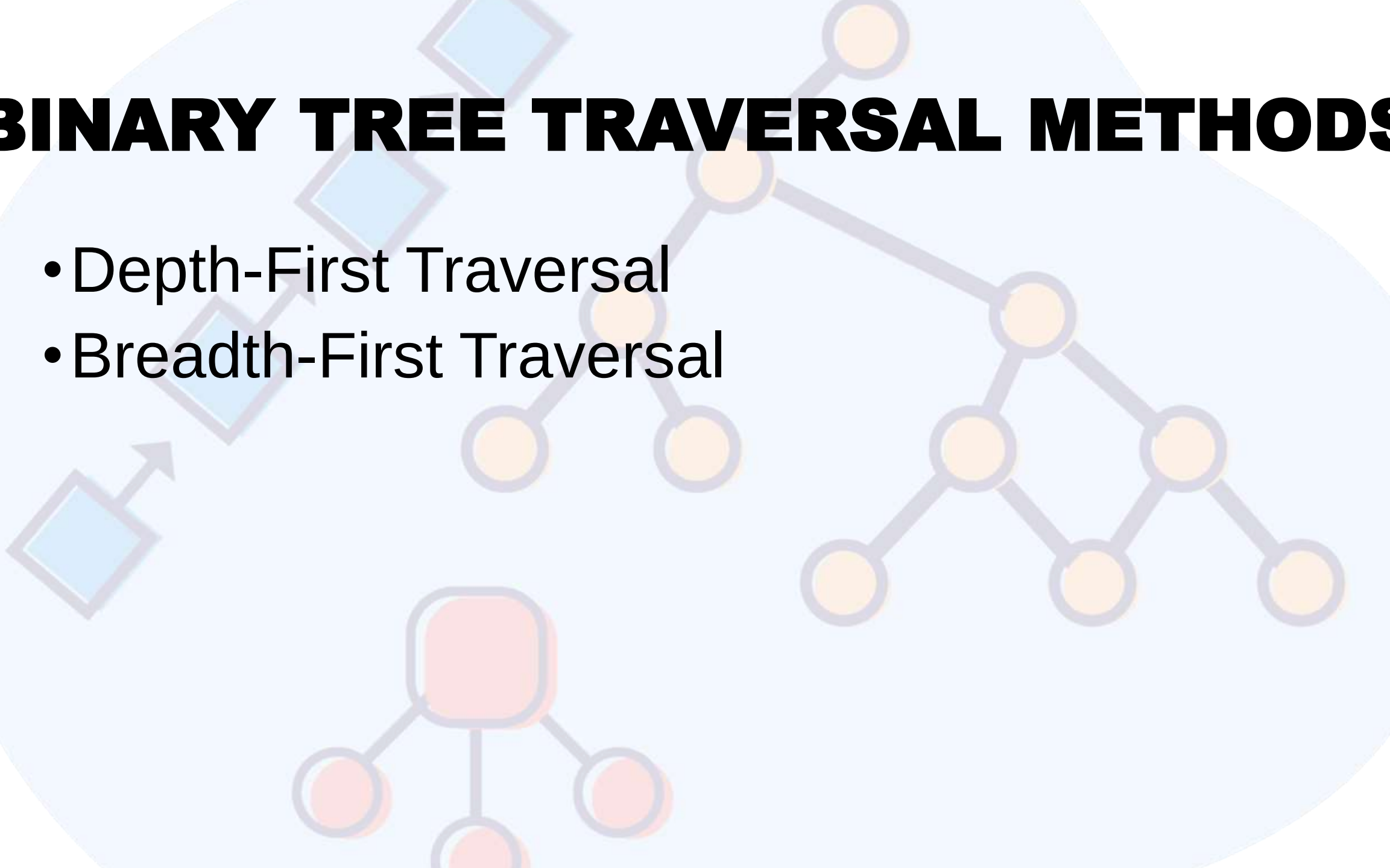
- The depth of a node is the number of edges from the node to the tree root node.

- **Height**

- The height of a node is the number of edges on the longest path from the node to a leaf

BINARY TREE TRAVERSAL METHODS

- Depth-First Traversal
- Breadth-First Traversal



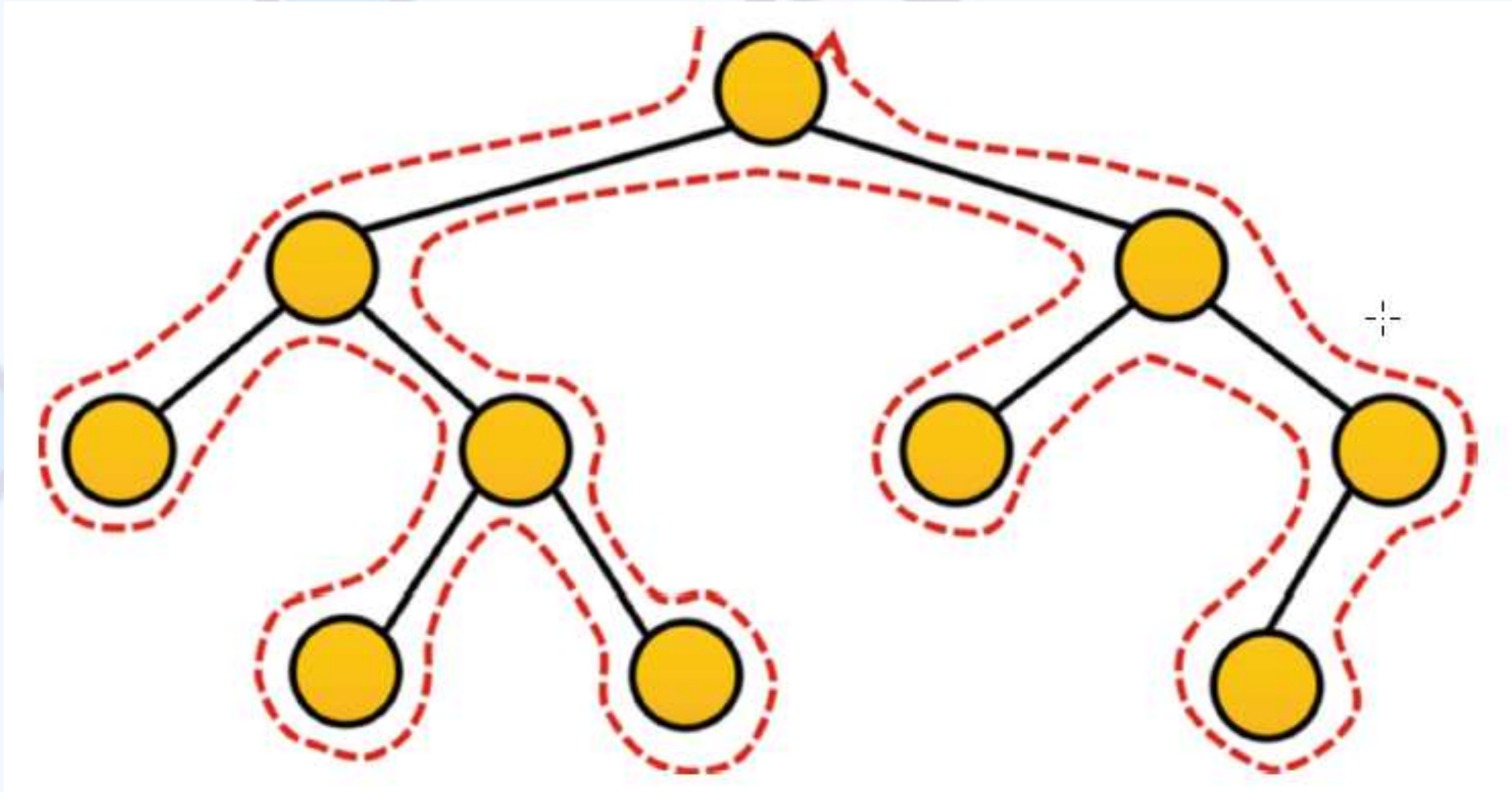
DEPTH-FIRST TRAVERSAL

- It explores as deep as possible down each branch before backtracking
- **Pre-order Traversal (Root-Left-Right):**
 - Visits the root node before its subtrees
 - This method is advantageous for cloning a tree or evaluating prefix expressions
- **In-Order Traversal (Left-Root-Right):**
 - Processes the left subtree, the root, and then the right subtree
 - It is especially useful for binary search trees to retrieve data in sorted order
- **Post-Order Traversal (Left-Right-Root):**
 - Accesses the current node after its subtrees
 - It is typically employed for deleting a tree or evaluating postfix expressions

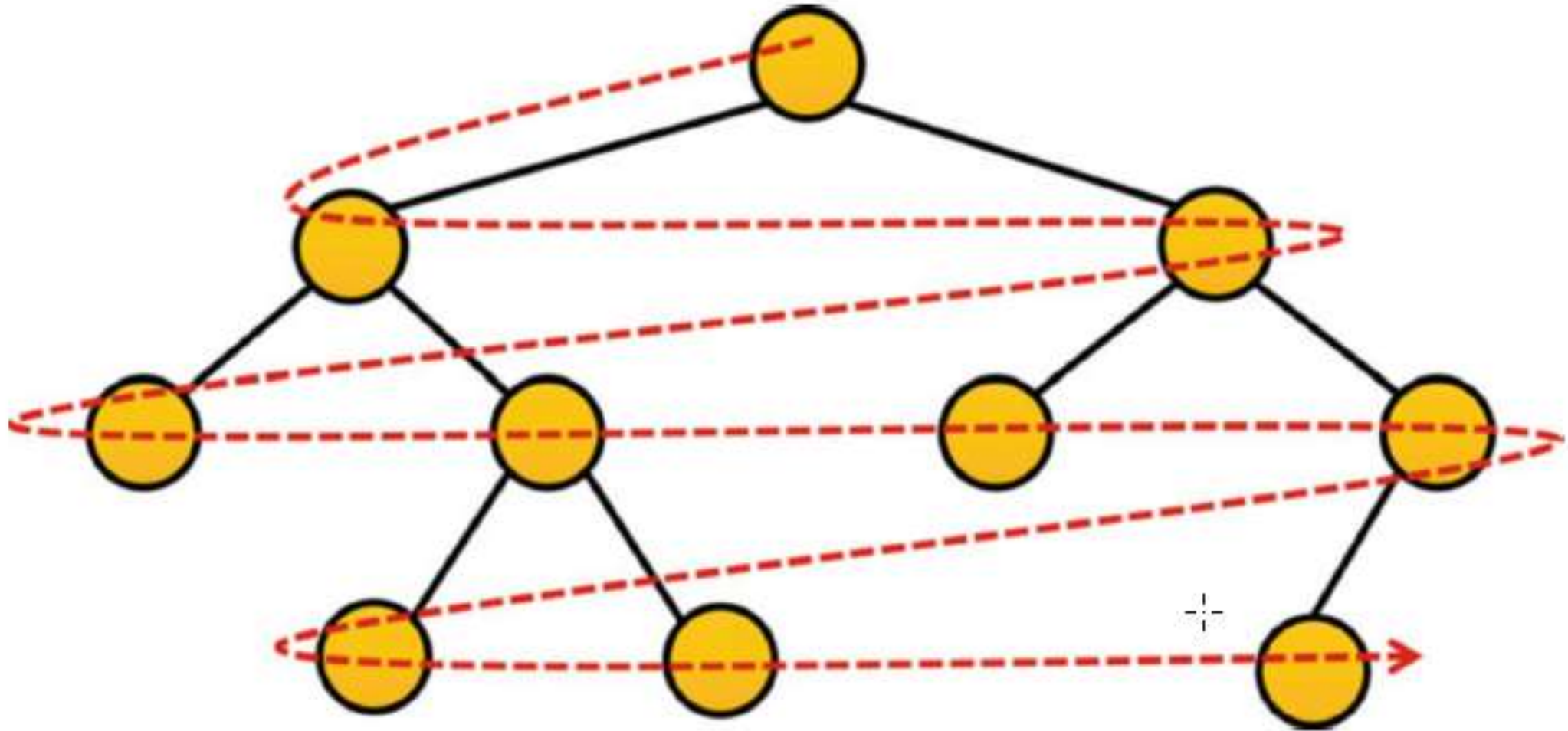
BREADTH-FIRST TRAVERSAL

- **Breadth-First Traversal, or Level-Order Traversal**
- It visits nodes across each level before proceeding to the next
- This approach is executed using a queue
- It is ideal when all nodes at one level must be processed before moving to the next

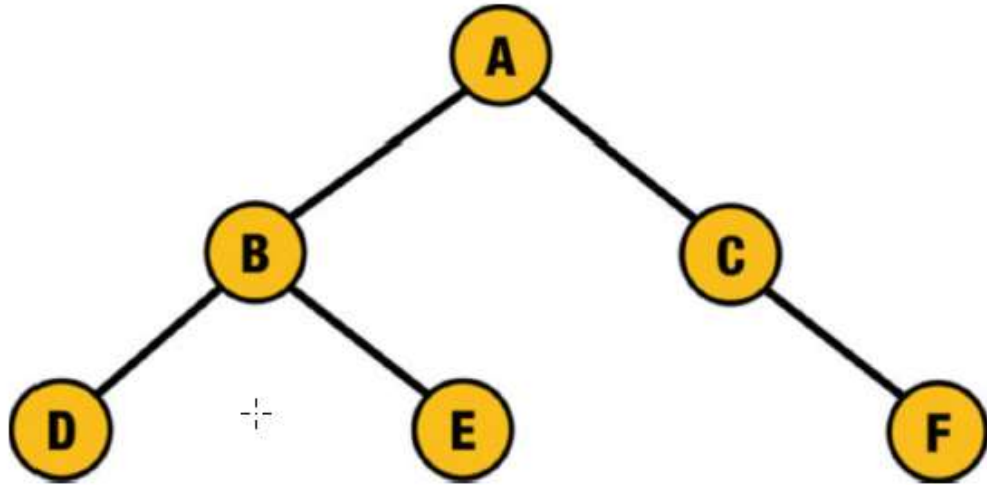
DEPTH-FIRST TRAVERSAL



BREADTH-FIRST TRAVERSAL



TASKS



- Find Pre Order Traversal
 - A, B, D, E, C, F
- Find In Order Traversal
 - D, B, E, A, C, F
- Find Post Order Traversal
 - D, E, B, F, C, A
- Find Level Order Traversal
 - A, B, C, D, E, F

COMPARISON OF TRAVERSAL METHODS

Traversal Type	Order of Visitation	Typical Applications
In-Order	Left subtree, Node, Right subtree	Retrieving data in sorted order in binary search trees
Pre-order	Node, Left subtree, Right subtree	Cloning trees, evaluating prefix expressions
Post-Order	Left subtree, Right subtree, Node	Deleting trees, evaluating postfix expressions
Level-Order	Across each level, from Left to Right	Processing nodes level by level, as in Breadth-First Search algorithms